

Manish Barti

GitHub: [su7ox](#) | LinkedIn: [manish-barti](#) | bartimanish36@gmail.com

SUMMARY

CS undergraduate building production-grade AI systems - autonomous agents, RAG pipelines, and high-throughput APIs (20K+ req/min).

EDUCATION

Graphic Era Deemed to be University

B.Tech, Computer Science Engineering

Dehradun, India

2023 – 2026

EXPERIENCE

Guidewire DEVTrials Hackathon 2026 | *Participant*

2026

- Built GigInsurance, an end-to-end parametric income insurance platform for gig delivery workers, under hackathon constraints spanning agentic AI, RAG pipelines, WhatsApp integration, and a payout engine.
- Designed and implemented a LangGraph-powered autonomous claim agent integrating 6 real-time verification tools (weather APIs, government feeds, policy RAG) to deliver claim decisions in **under 30 seconds** without human intervention.
- Engineered a full async backend (FastAPI + PostgreSQL) with OTP-based WhatsApp session auth, Alembic migrations, and a provider-agnostic messaging layer supporting Twilio API.
- Architected and deployed a complete insurance technology stack - parametric trigger engine, XGBoost risk scoring, Isolation Forest fraud detection, and ChromaDB RAG independently within the hackathon timeline.

PROJECTS

SafeNav - Web & Link Safety Analysis Platform | *Redis, asyncio, PostgreSQL* [Demo](#) | [GitHub](#)

- Architected a multi-layer threat pipeline (Lexical, SSL/TLS, IP Intelligence, Content) with compound signal scoring, replacing black-box verdicts with auditable per-rule risk breakdowns.
- Cut scan latency to **under 3s** by parallelizing 5 network modules via `asyncio.gather` with Redis-backed caching, eliminating redundant queries on repeated scans.
- Bootstrapped a **100K-domain** in-memory trust registry at startup, reducing false positives on CDN-hosted sites (AWS, Azure, Cloudflare) with zero per-request overhead.
- Trained a Random Forest classifier (scikit-learn, 300 estimators) on **650K+** labeled URLs with 17 lexical features and deployed as a FastAPI singleton, reusing pre-computed `LexicalReport` objects to eliminate train-serve skew.

APIGate – Distributed API Gateway | *FastAPI, Redis, Docker, Grafana*

[GitHub](#)

- Built a reverse-proxy API Gateway with Redis-backed sliding-window rate limiting, JWT authentication middleware, and request routing across multiple containerized backend services.
- Implemented a circuit breaker pattern with open/half-open/closed states to prevent cascade failures across downstream microservices, reducing error propagation under high load.
- Instrumented real-time observability with Prometheus metrics and Grafana dashboards tracking p99 latency, request throughput, and error rates across all proxied services.
- Containerized the full stack (gateway, Redis, Prometheus, Grafana, mock backends) with Docker Compose, enabling one-command local deployment of the complete microservices environment.

TECHNICAL SKILLS

Programming:	C/C++, Python, SQL
Web Development:	HTML, CSS, JavaScript, React.js
Backend :	FastAPI, Express.js, REST APIs, JWT Auth
Databases:	PostgreSQL, MySQL, Redis, Vector DB
Generative AI/ML:	LangChain, LangGraph, RAG, NLP, XGBoost, Isolation Forest, Pandas, NumPy
Cloud & DevOps:	AWS EC2, AWS S3, Azure, Docker, Linux, Prometheus, Grafana, Kubernetes
Tools:	GitHub, Postman, VS Code, PyCharm

CERTIFICATIONS

AWS Cloud Essentials AWS Skill Builder

Jan – June 2026

AWS AI Practitioner AWS Skill Builder

Jul – Nov 2025

LEADERSHIP & INVOLVEMENT

Member, Google Developer Student Clubs (GDSC)

2023 – 2024